



ITA-ECBS: A Bounded-Suboptimal Algorithm for the Combined Target-Assignment and Path-Finding Problem

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Multi-Agent Path Finding Problem (MAPF)





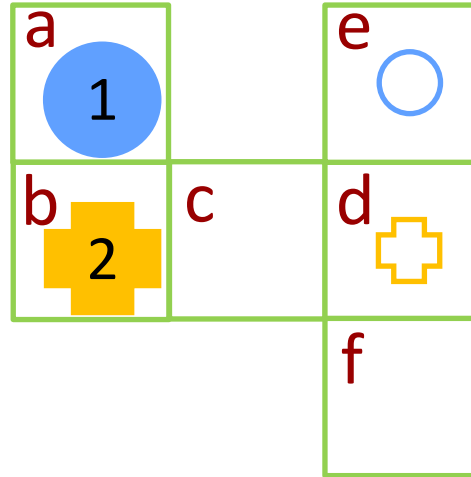
Multi-Agent Path Finding Problem (MAPF)





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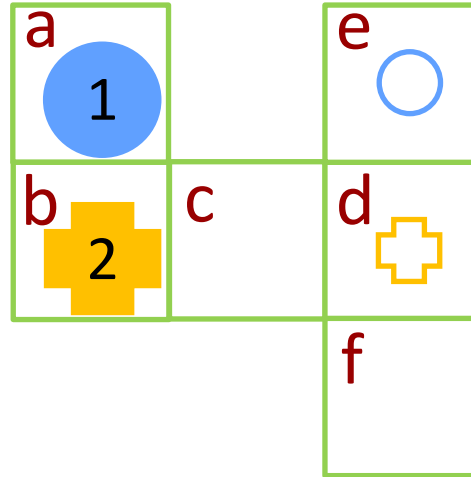
- Input:
 - a known graph
 - multiple agents (each with one start and one target location)



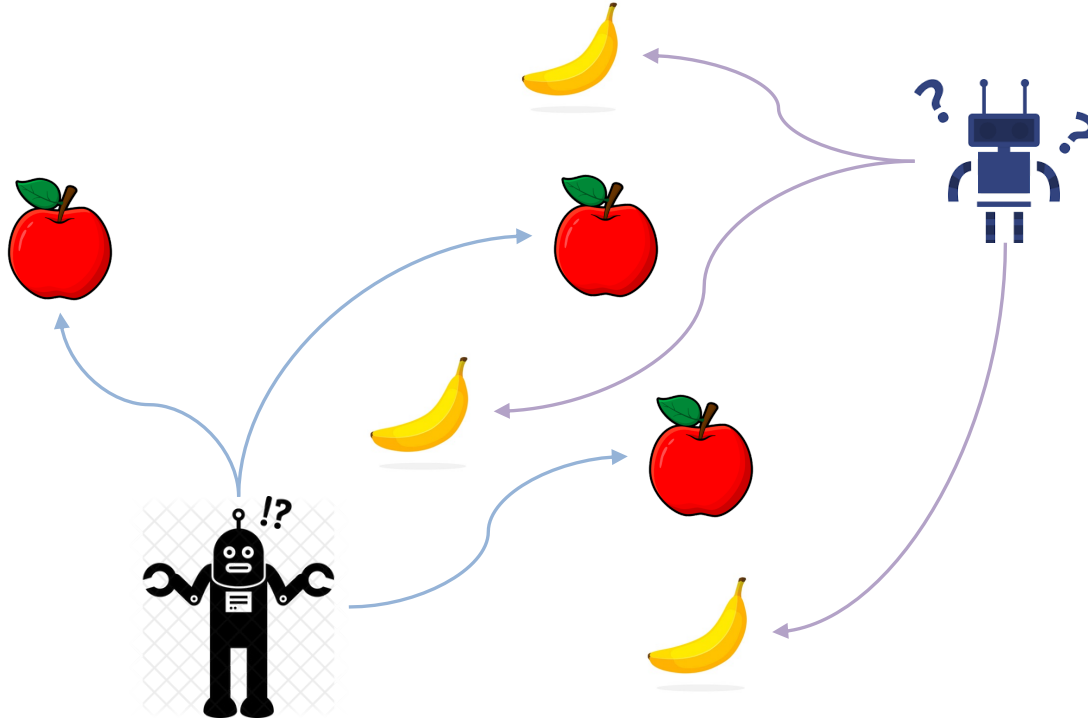


Multi Agent Path Finding Problem (MAPF)

- Output:
 - Collision-free paths for all agents that minimize flowtime
 - Flowtime: sum of all agents arrival time



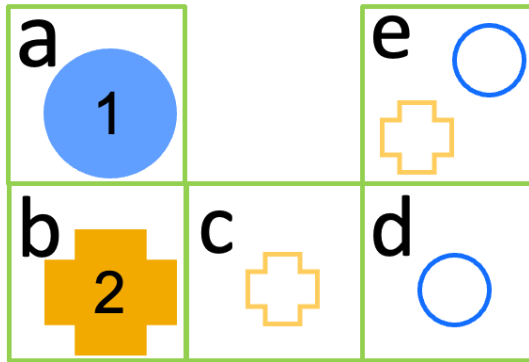
Target-Assignment and Path-Finding Problem (TAPF)





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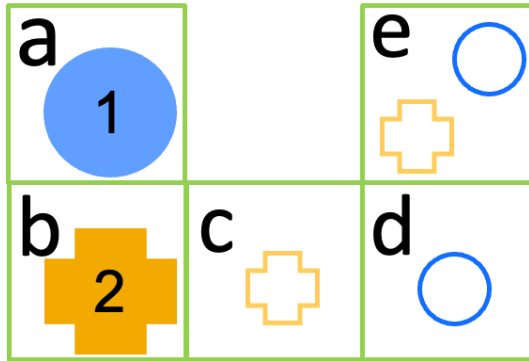
- Input:
 - a known graph
 - agents (each with a start location and a target location set)





Target-Assignment and Path-Finding Problem (TAPF)

- Output:
 - Agent only needs go to **one target** in the set
 - **Collision-free** paths for all agents and **minimize flowtime**





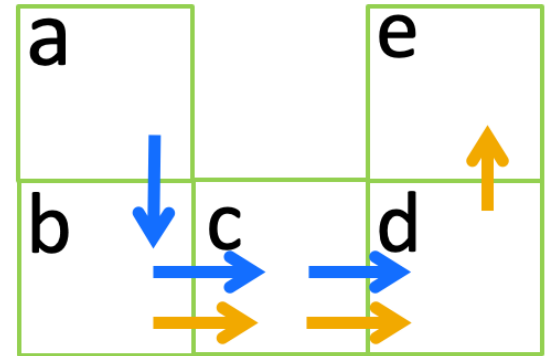
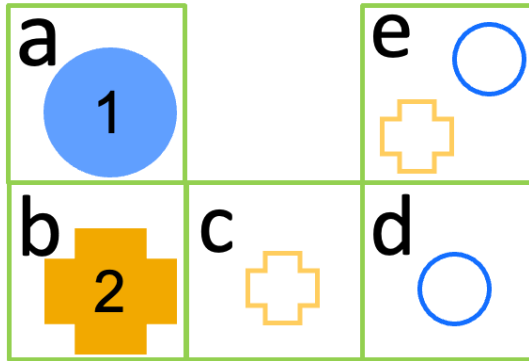
Target-Assignment and Path-Finding Problem (TAPF)

- Output:
 - Agent only needs go to **one target** in the set
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Target Assignment (TA):

1 → d

2 → e



Incremental Target Assignment Conflict-Based Search (ITA-CBS)

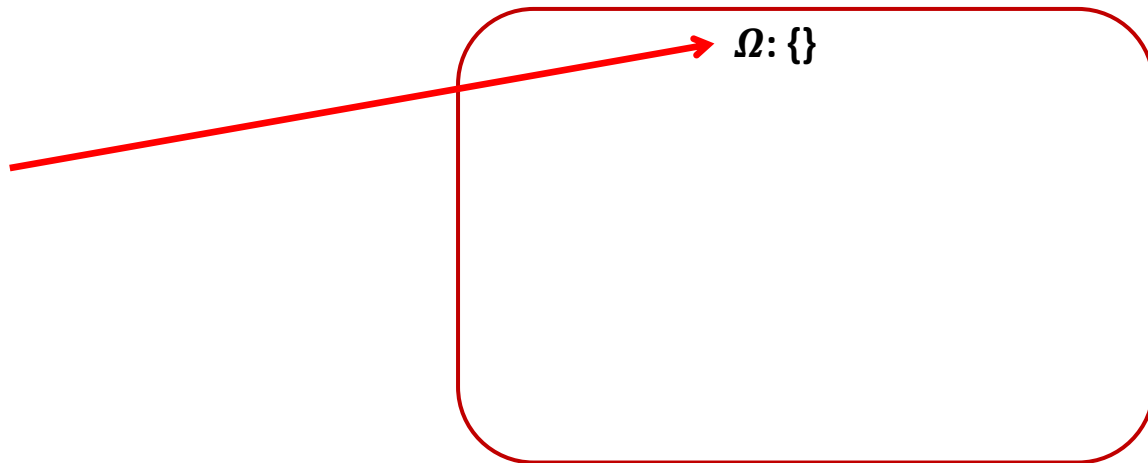


- Optimal SOTA method for TAPF

Incremental Target Assignment Conflict-Based Search (ITA-CBS)



- Optimal SOTA method for TAPF
- ITA-CBS Tree Node:
 - Constraint Set Ω



Incremental Target Assignment Conflict-Based Search (ITA-CBS)



- Optimal SOTA method for TAPF
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 - Shortest Path Cost Matrix M under Ω

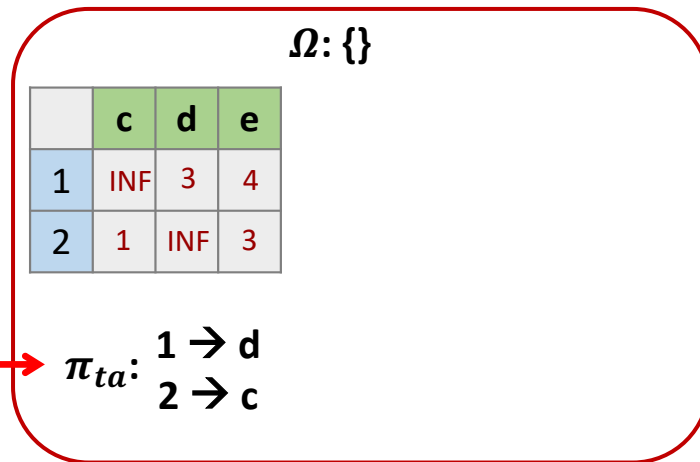
$\Omega: \{\}$

	c	d	e
1	INF	3	4
2	1	INF	3

Incremental Target Assignment Conflict-Based Search (ITA-CBS)



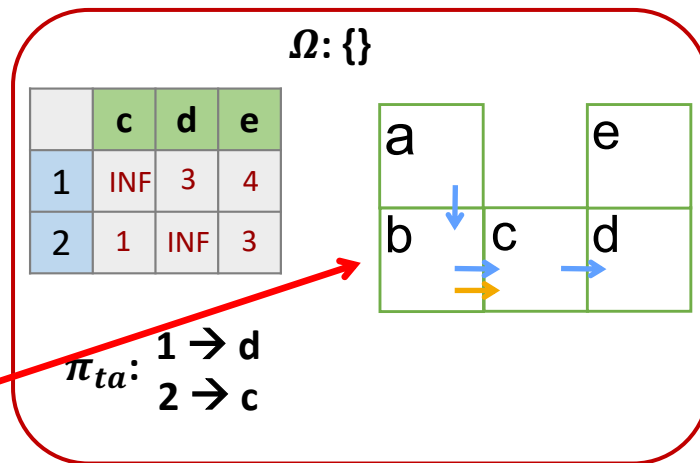
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 - Optimal Target Assignment from M



Incremental Target Assignment Conflict-Based Search (ITA-CBS)



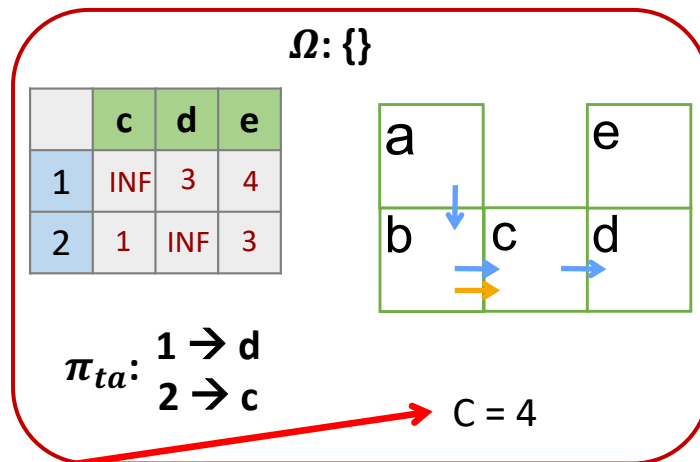
- Optimal SOTA method for TAPF
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 - Solution π based on TA



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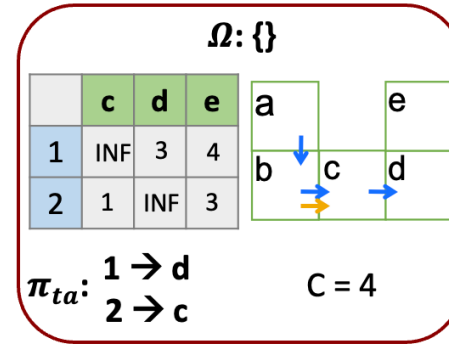
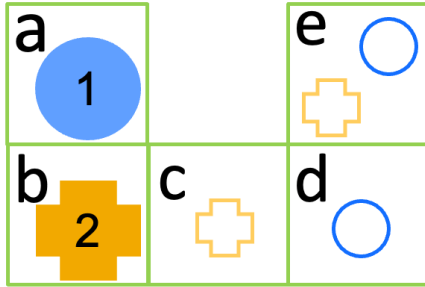
- Optimal SOTA method for TAPF
- ITA-CBS Tree Node:
 - Constraint Set Ω
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 - Solution π based on TA
 - Flowtime $\mathcal{C}$





Incremental Target Assignment Conflict-Based Search (ITA-CBS)

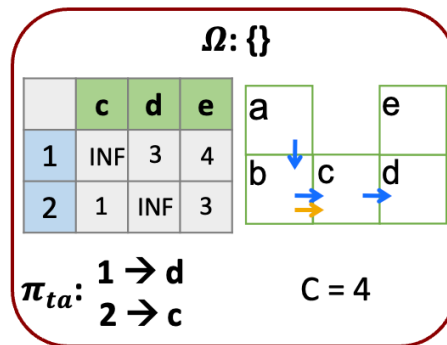
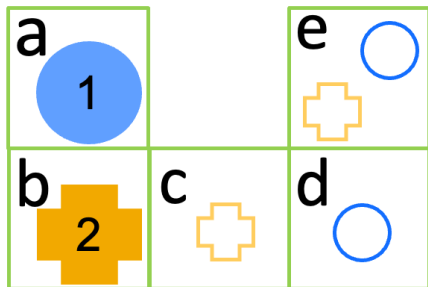
- Optimal SOTA method for TAPF



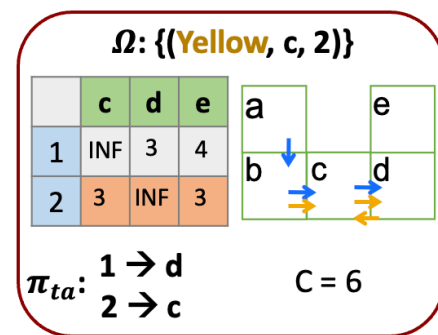
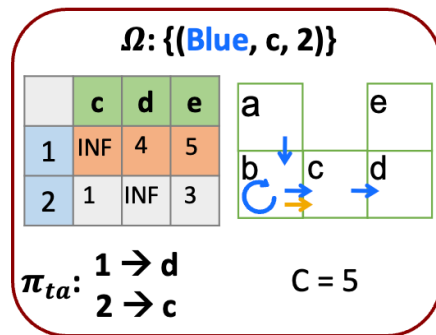


Incremental Target Assignment Conflict-Based Search (ITA-CBS)

- Optimal SOTA method for TAPF



Conflict at (c, time: 2)



Incremental Target Assignment Conflict-Based Search (ITA-CBS)



- Optimal SOTA method for TAPF
- $TAPF \geq MAPF$, and MAPF is NP-hard

Incremental Target Assignment Conflict-Based Search (ITA-CBS)

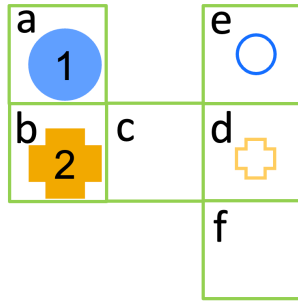


- Optimal SOTA method for TAPF
- $TAPF \geq MAPF$, and MAPF is NP-hard
- All optimal methods have huge time complexity

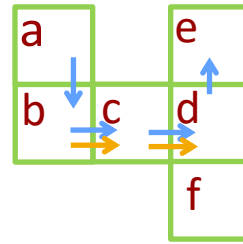


Conflict-Based Search (CBS)

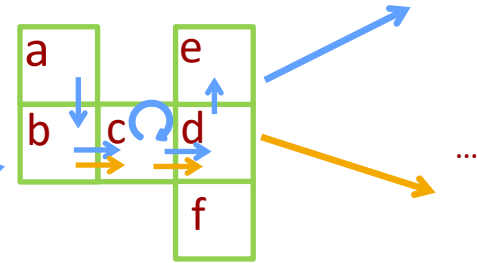
- Optimal method for MAPF



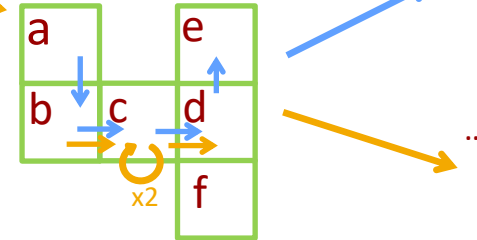
Conflict at location **d** and timestep **3**



Agent 1 can't be
at location d at
timestep 3



Agent 2 can't be
at location d at
timestep 3





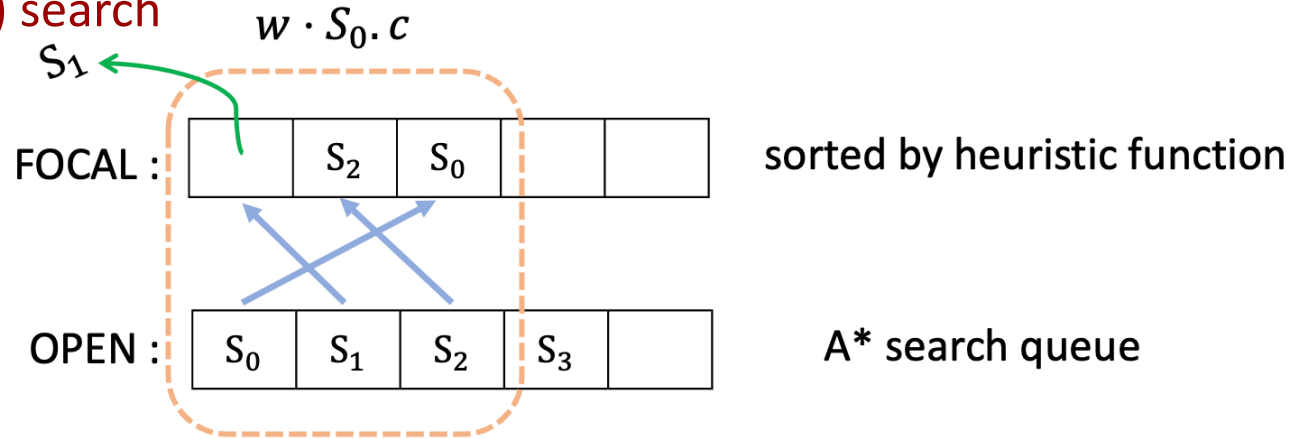
Enhanced CBS (ECBS)

It can find a valid solution with cost $c \leq w * c_{opt}$



Enhanced CBS (ECBS)

Low-Level (path) search



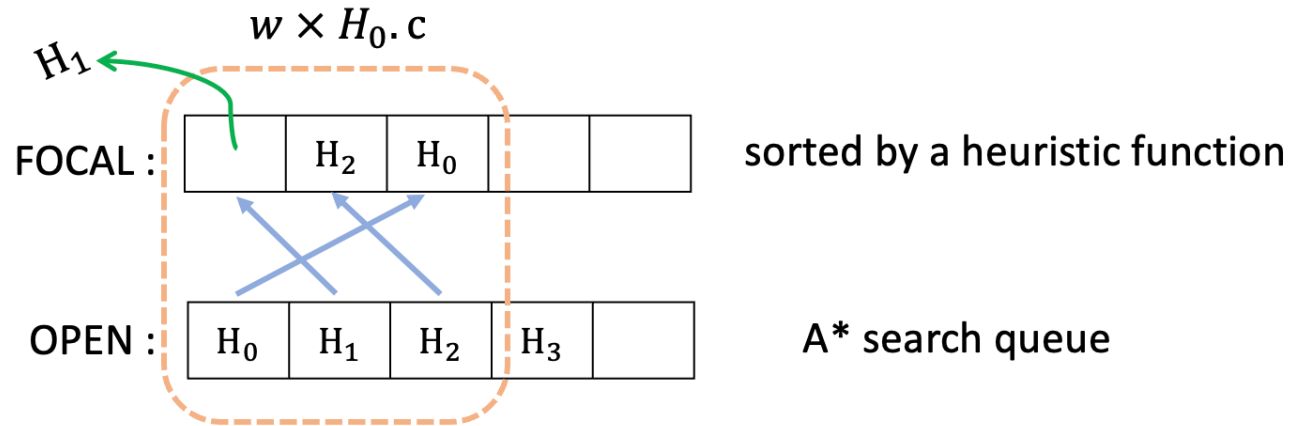
LB, path = FocalSearch(start location, target location)

$$\text{Path}.c \leq w \cdot LB \leq w \cdot c_{opt}$$



Enhanced CBS (ECBS)

High-Level (CBS node) search

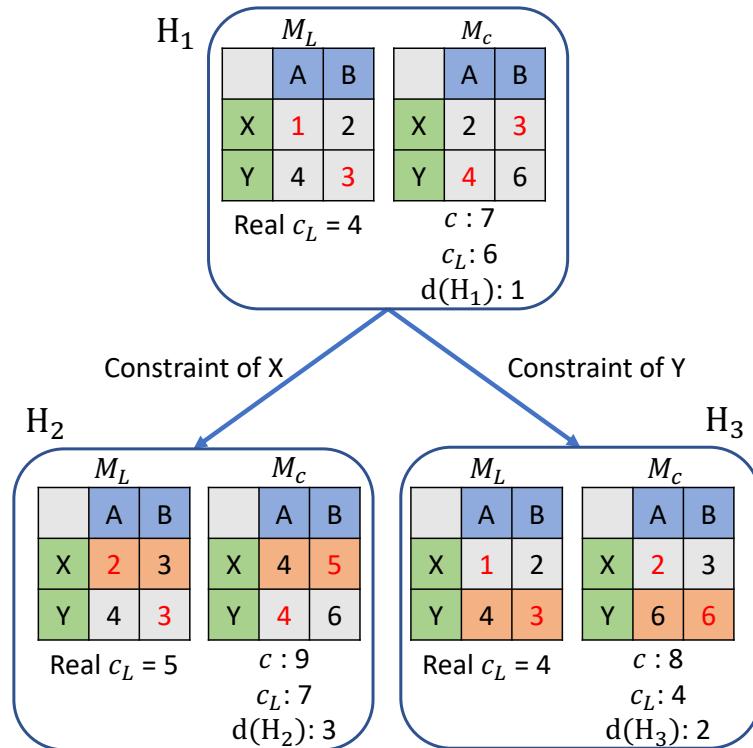


$$H.c_L = \sum path.LB$$

Incremental Target Assignment Conflict-Based Search (ITA-CBS)



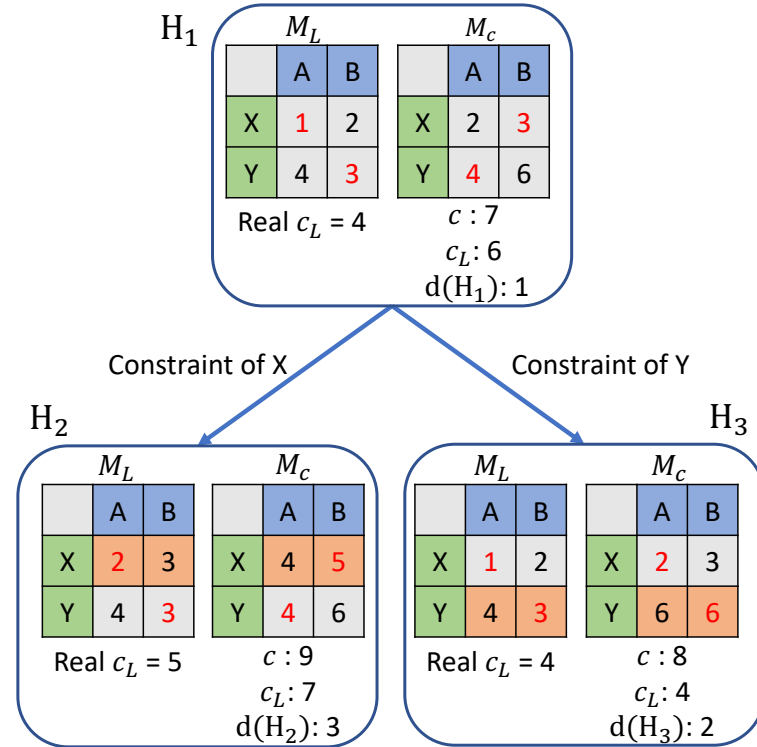
- If we **directly** apply ECBS method to ITA-CBS.....





Incremental Target Assignment Conflict-Based Search (ITA-CBS)

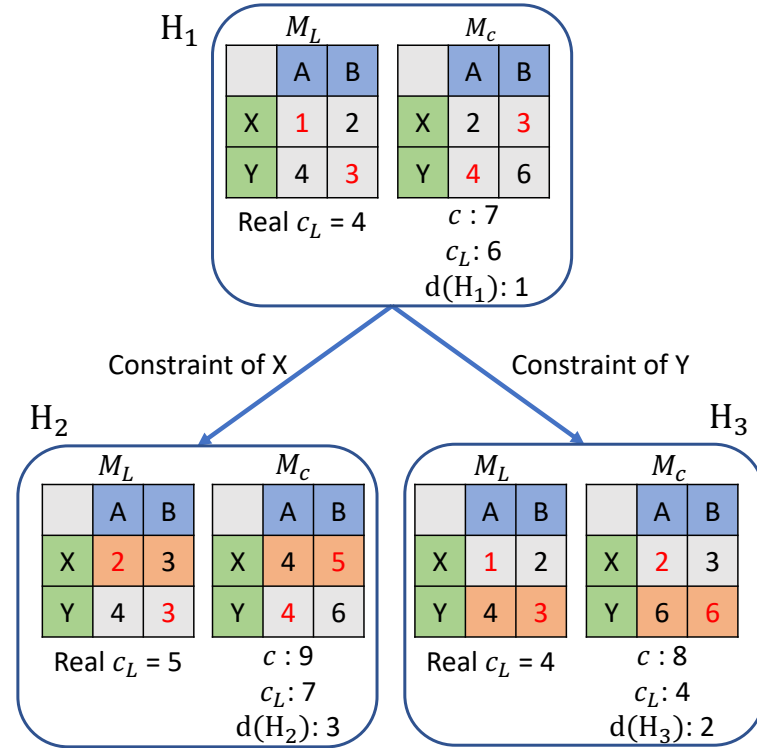
- If we directly apply ECBS method to ITA-CBS.....
- The optimal TA solution of M_L is **misaligned** with TA of M_C





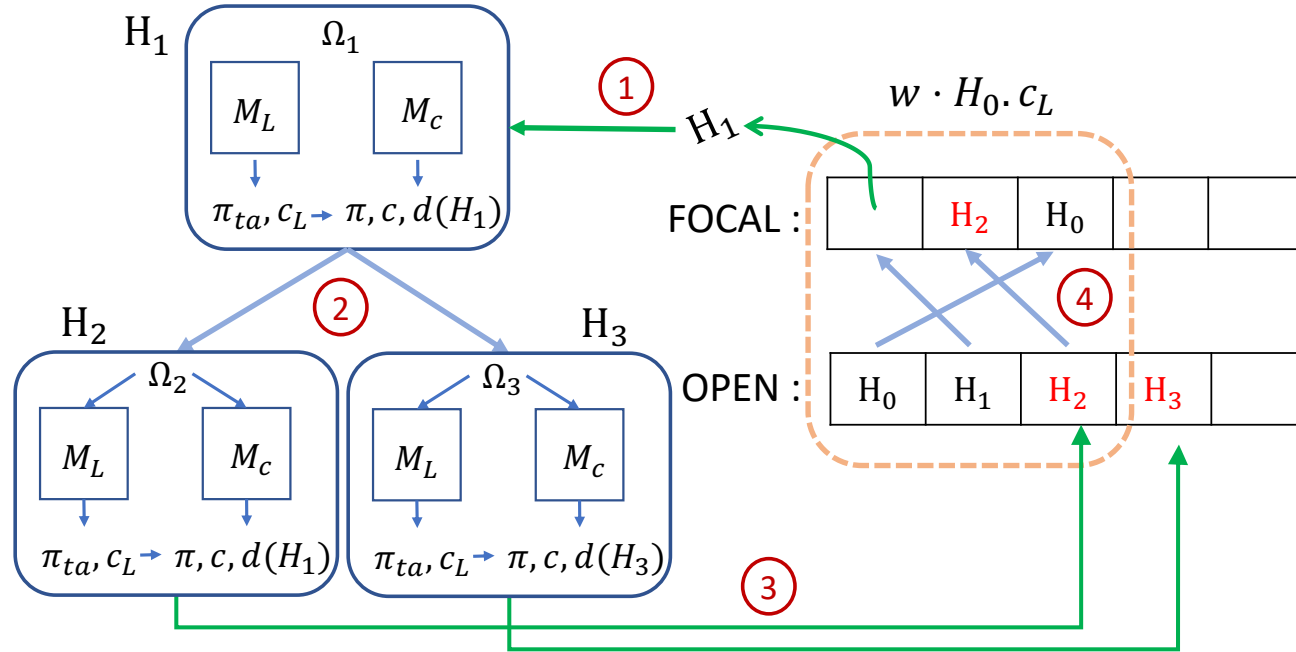
Incremental Target Assignment Conflict-Based Search (ITA-CBS)

- If we directly apply ECBS method to ITA-CBS.....
- The optimal TA solution of M_L is misaligned with TA of M_C
- The final collision-free solution may **not a bounded-suboptimal** solution

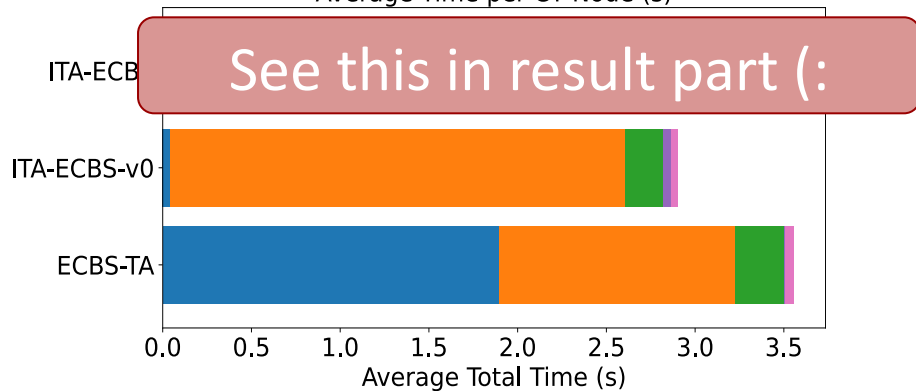
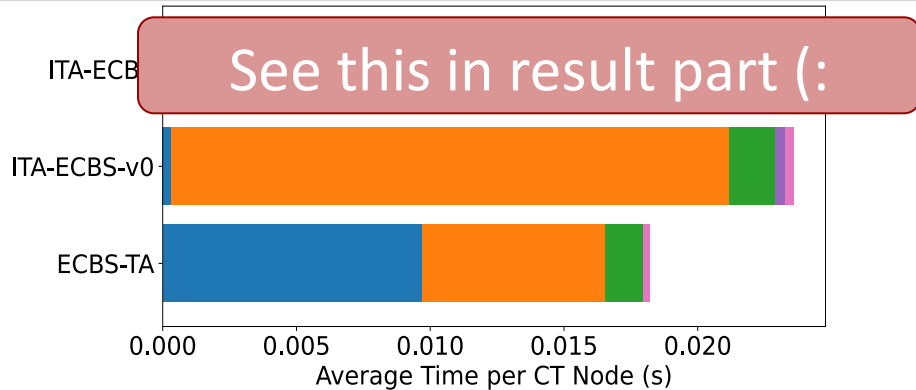
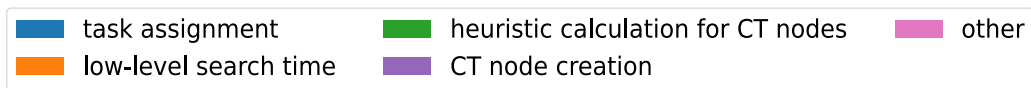




ITA-ECBS-v0 (ours)

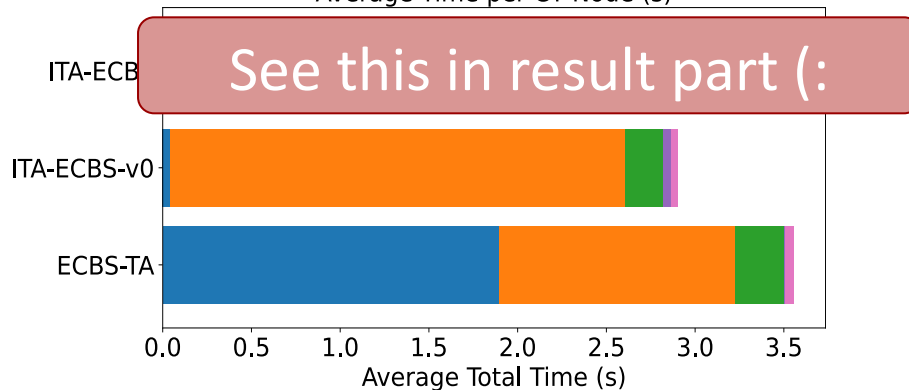
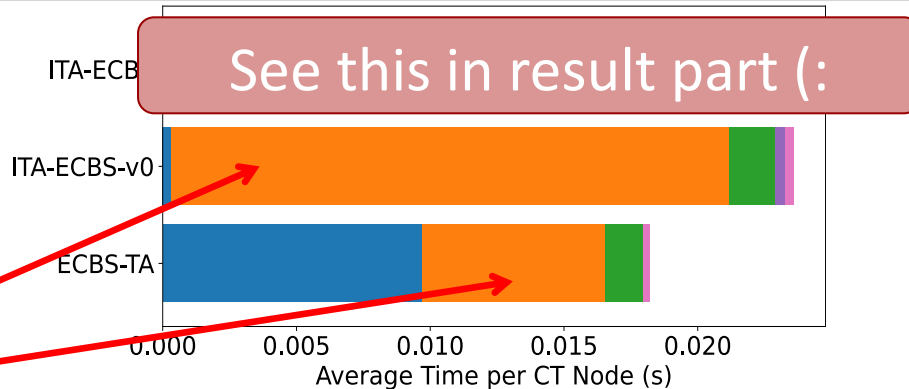
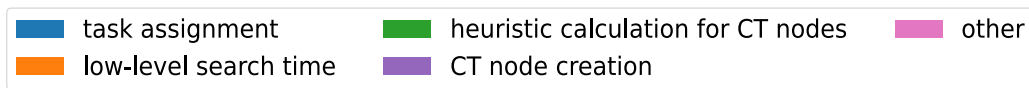


ITA-ECBS-v0 (ours)



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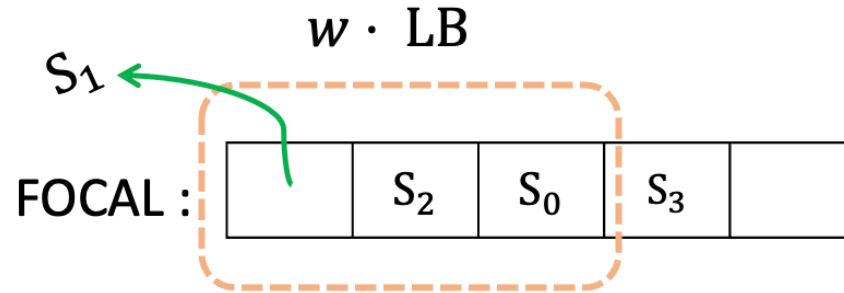
Average path search time
(per CT node) is longer
than ECBS-TA





ITA-ECBS (ours)

Low-level path search

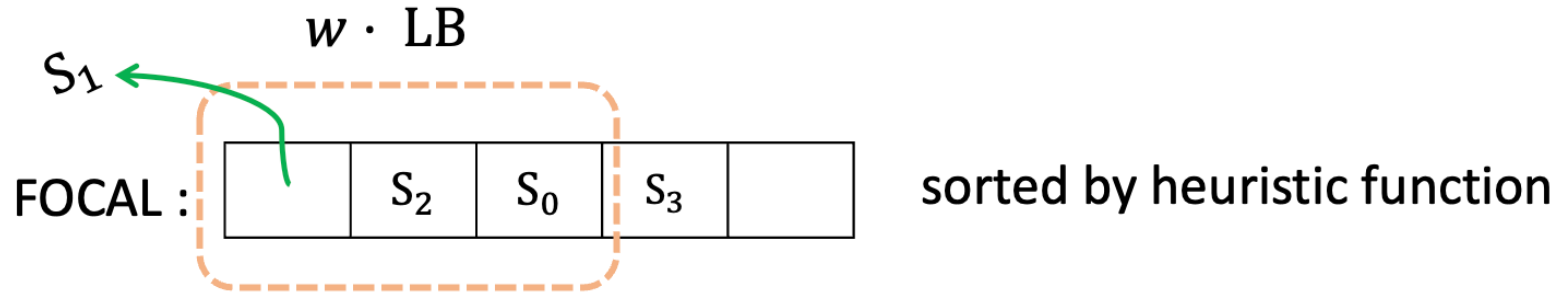


sorted by heuristic function



ITA-ECBS (ours)

Low-level path search



$$LB \leq \textit{shortest path cost}$$

ITA-ECBS (ours)



Low-level path search

```
function GETPATHCOSTANDB
  if ALGO_TYPE = ITA-ECBS-v0 then
     $cost, LB = \text{focalSearch}(G, s_k, g_x, Q.\Omega, w)$ 
  if ALGO_TYPE = ITA-ECBS then
     $LB = \text{shortestPathSearch}(G, s_k, g_x, \Omega_0)$ 
     $cost = \text{searchWithLB}(G, s_k, g_x, Q.\Omega, w, LB)$ 
  return  $cost, LB$ 
```


Experiment Results

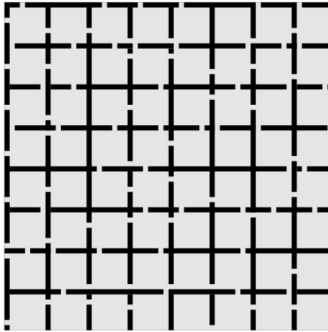




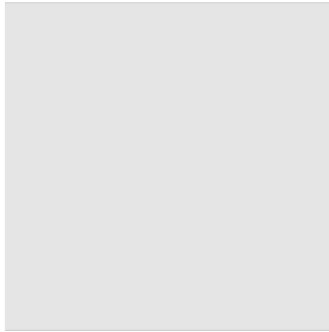
Experiment Results



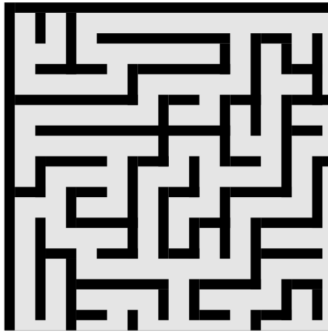
(a) den312d



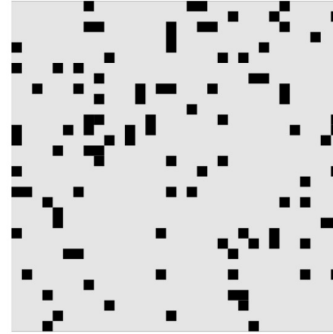
(b) room-64-64-8



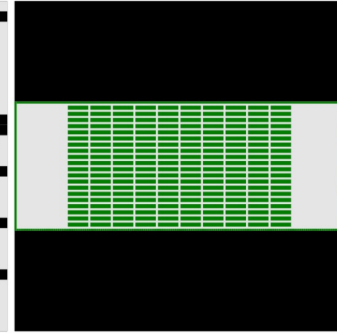
(c) empty-32-32



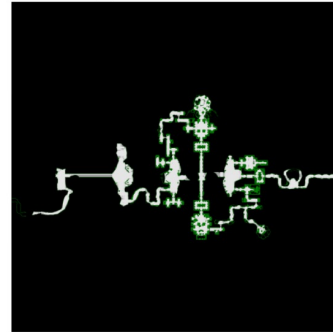
(d) maze-32-32-2



(a) random-32-32-10



(b) warehouse-10-20-10-2-1



(c) orz900d



(d) Boston-0-256

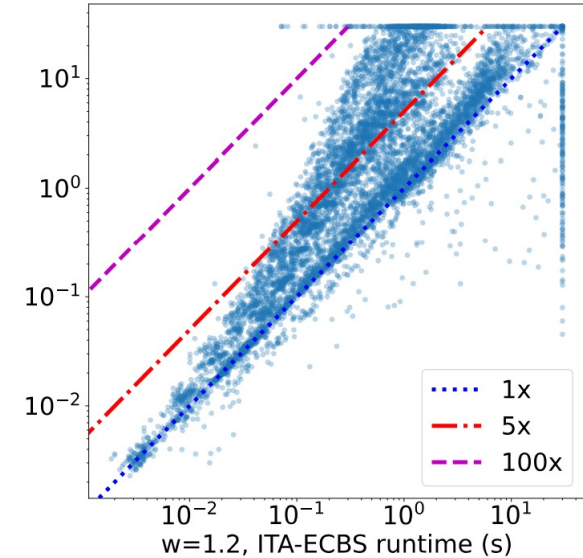
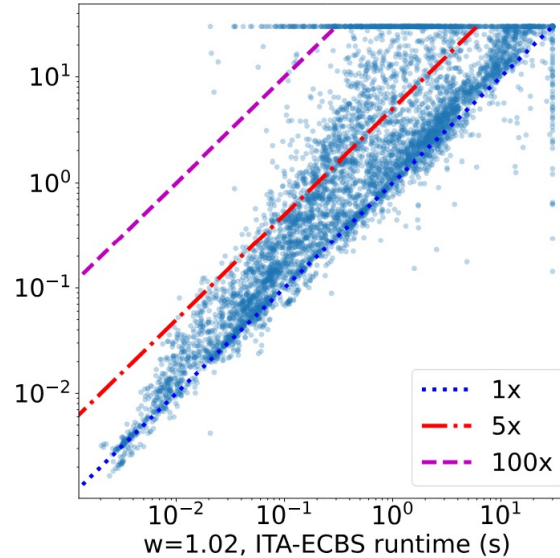
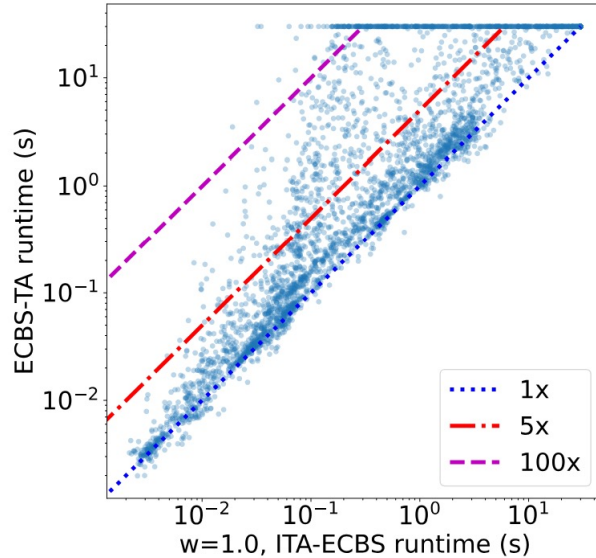


Experiment Results

- Agent: 10 — 150
- target set size: 5 — 30 (depends on map size)
- $w: \{ 1, 1.01, 1.02, 1.03, 1.04, 1.05, 1.1, 1.2 \}$
- $8(\#w) * [15(\#agent) * 8(\#map) * 4(\#ratio) * 20(\# \text{ test cases for one setting})] = 76800$



Experiment Results



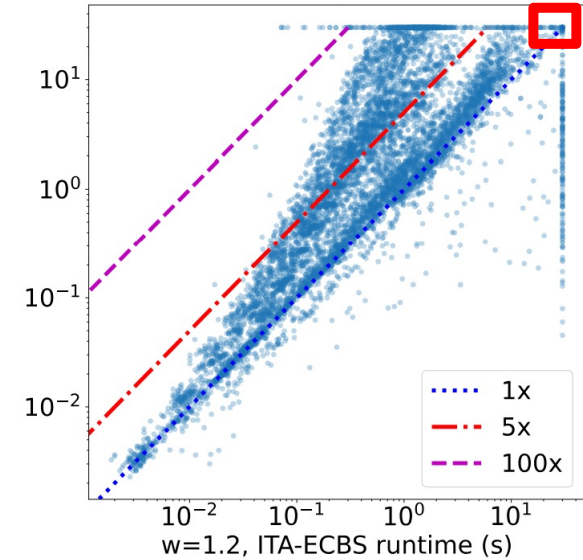
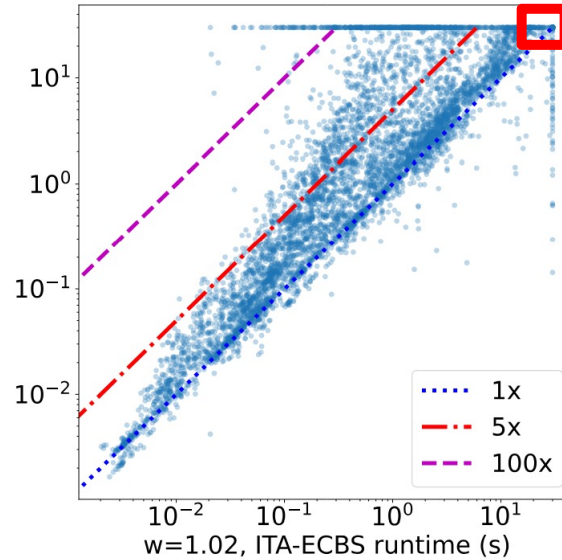
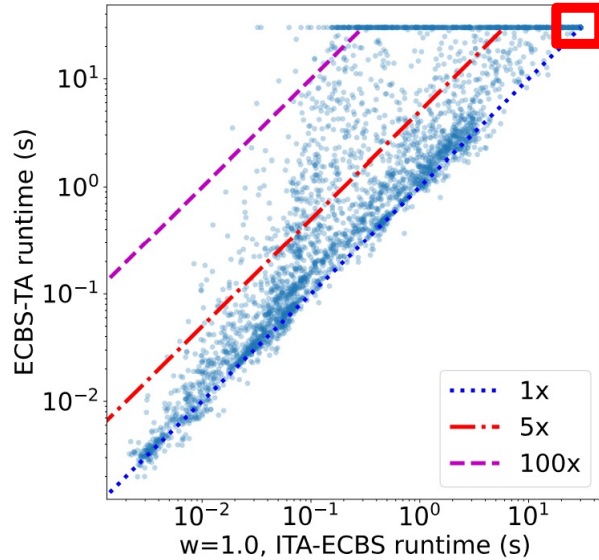
X-axis: Our method runtime

Y-axis: Baseline method runtime

Time limit: 30s



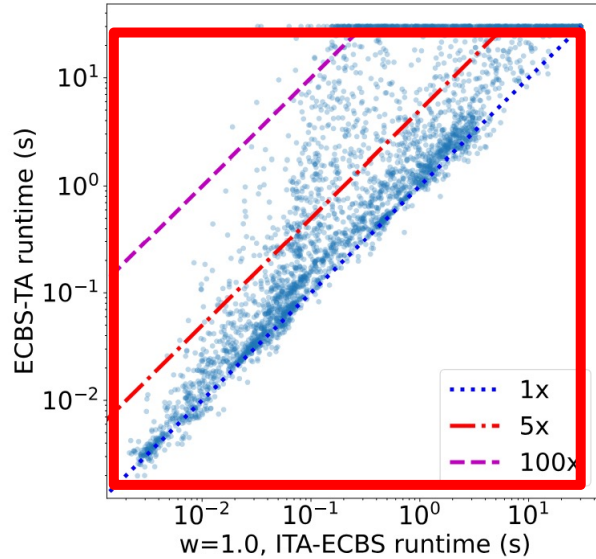
Experiment Results



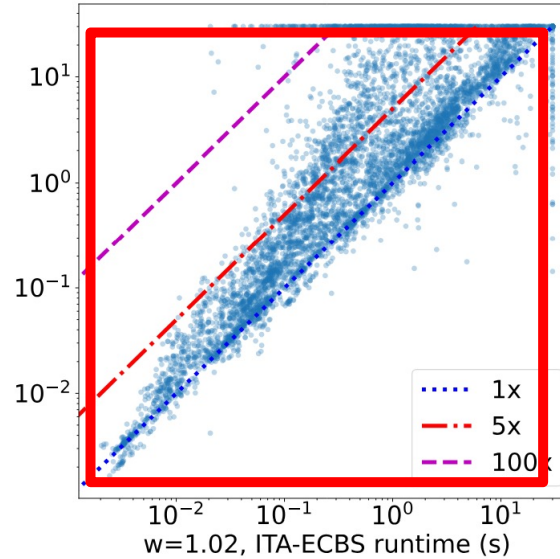
- 76800 testcases
- 22767 both failed



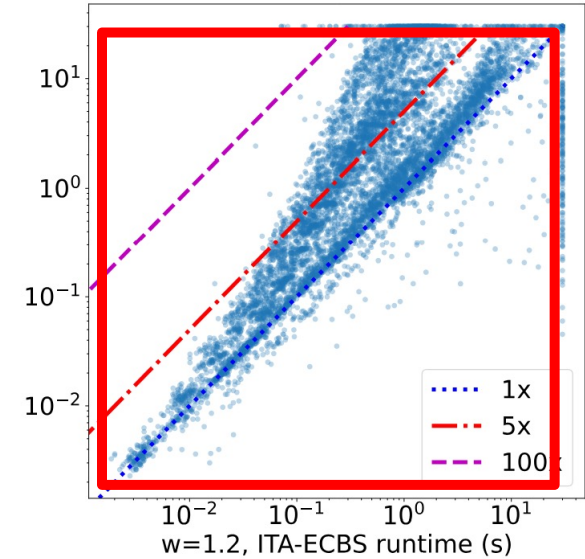
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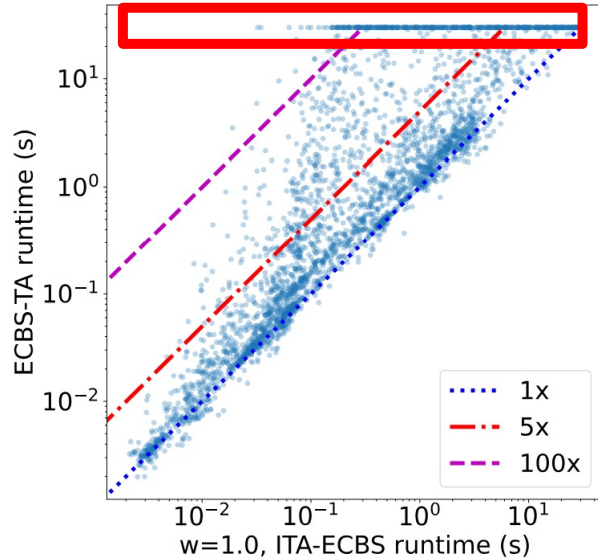


- 48334 both solved

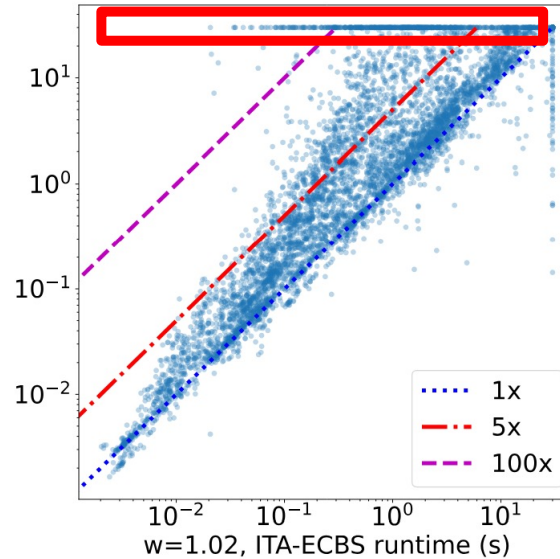




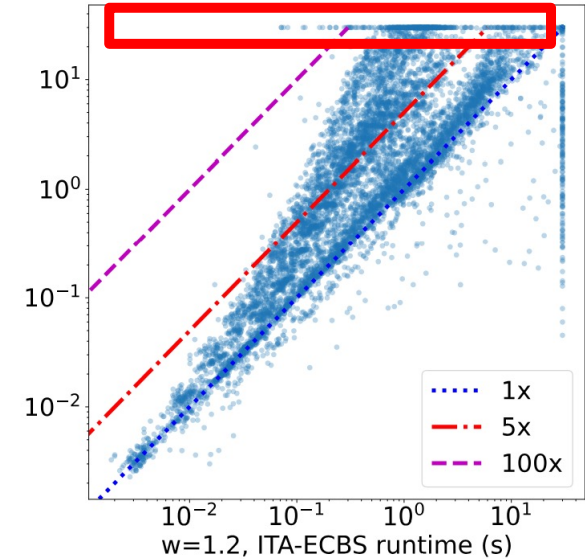
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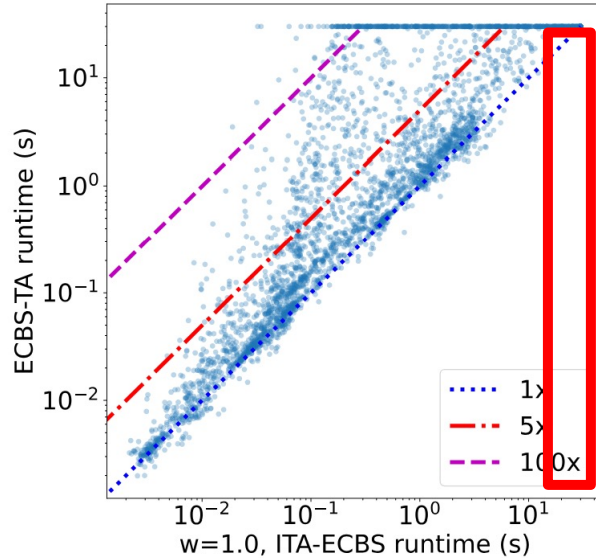


- 48334 both solved
- 5190 solved by ITA-ECBS only

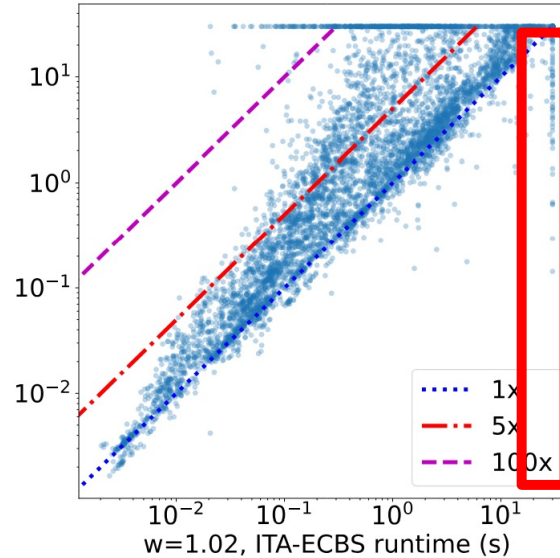




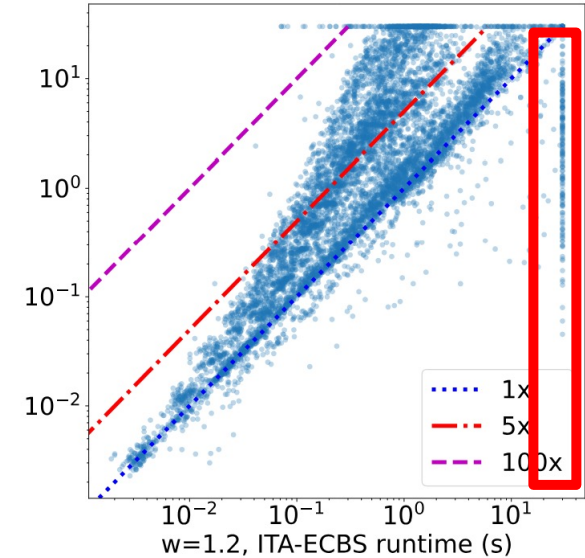
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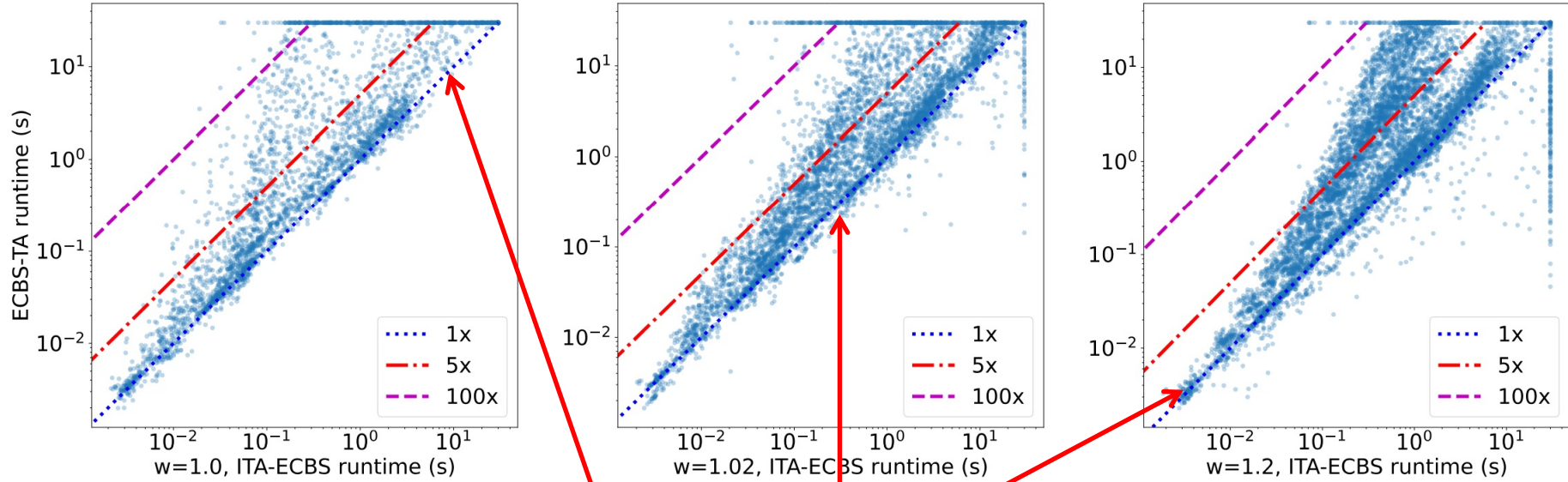


- 22767 both solved
- 5190 solved by ITA-ECBS only
- 509 solved by ECBS-TA only





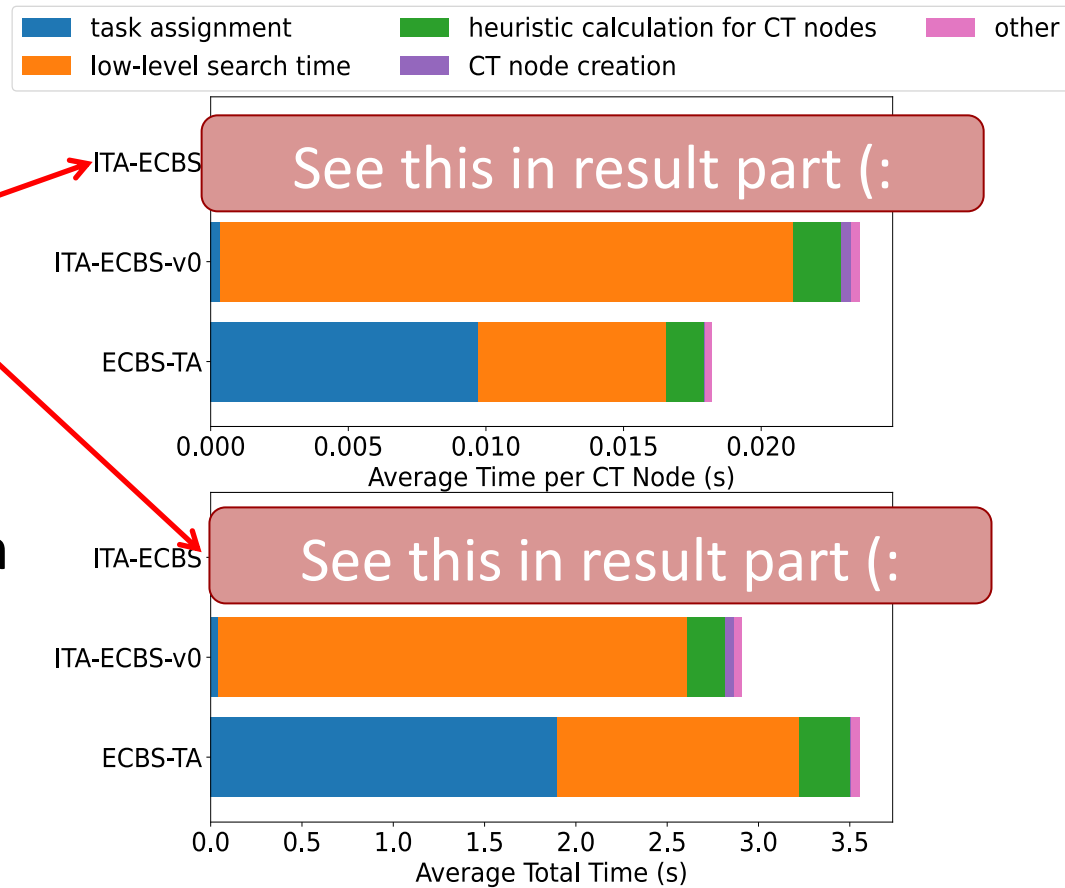
Experiment Results



- 87.42%: ITA-ECBS faster than ECBS-TA

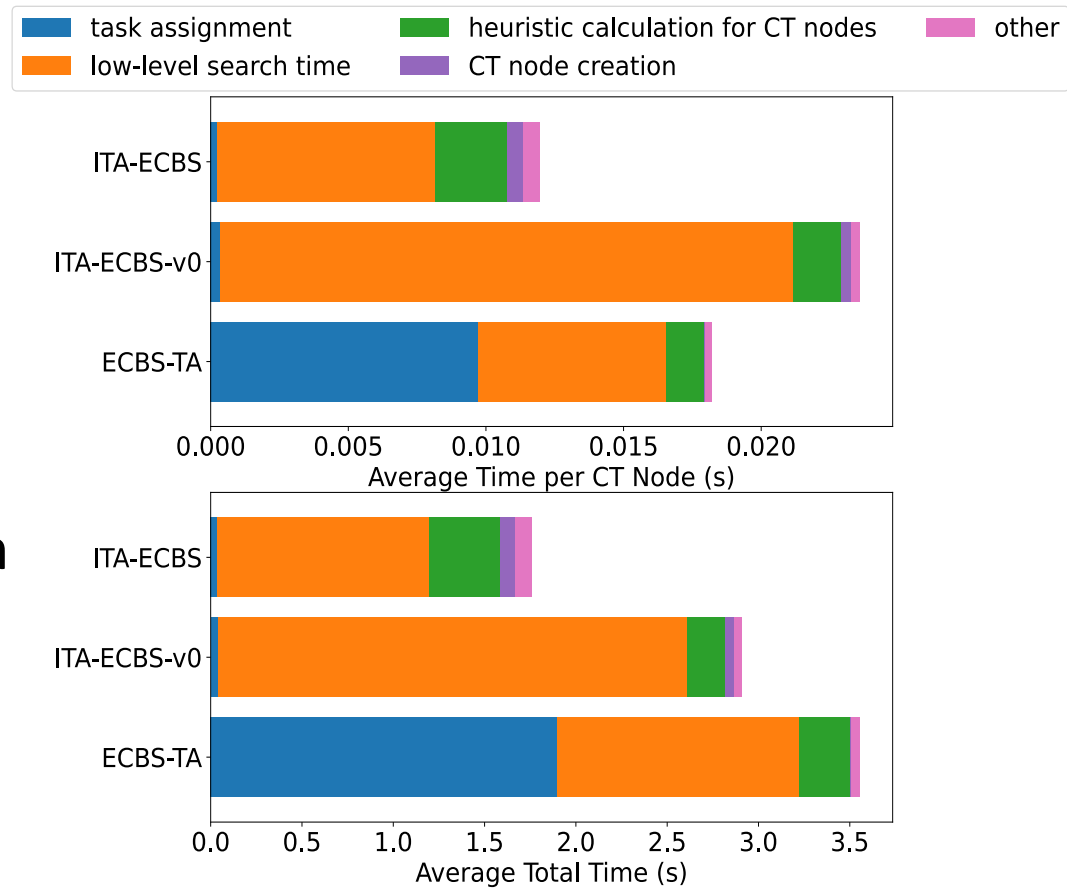
Experiment Results

- ITA-ECBS: two searches (shortest path search + focal search)
- ITA-ECBS-v0: one focal search
- ECBS-TA: baseline



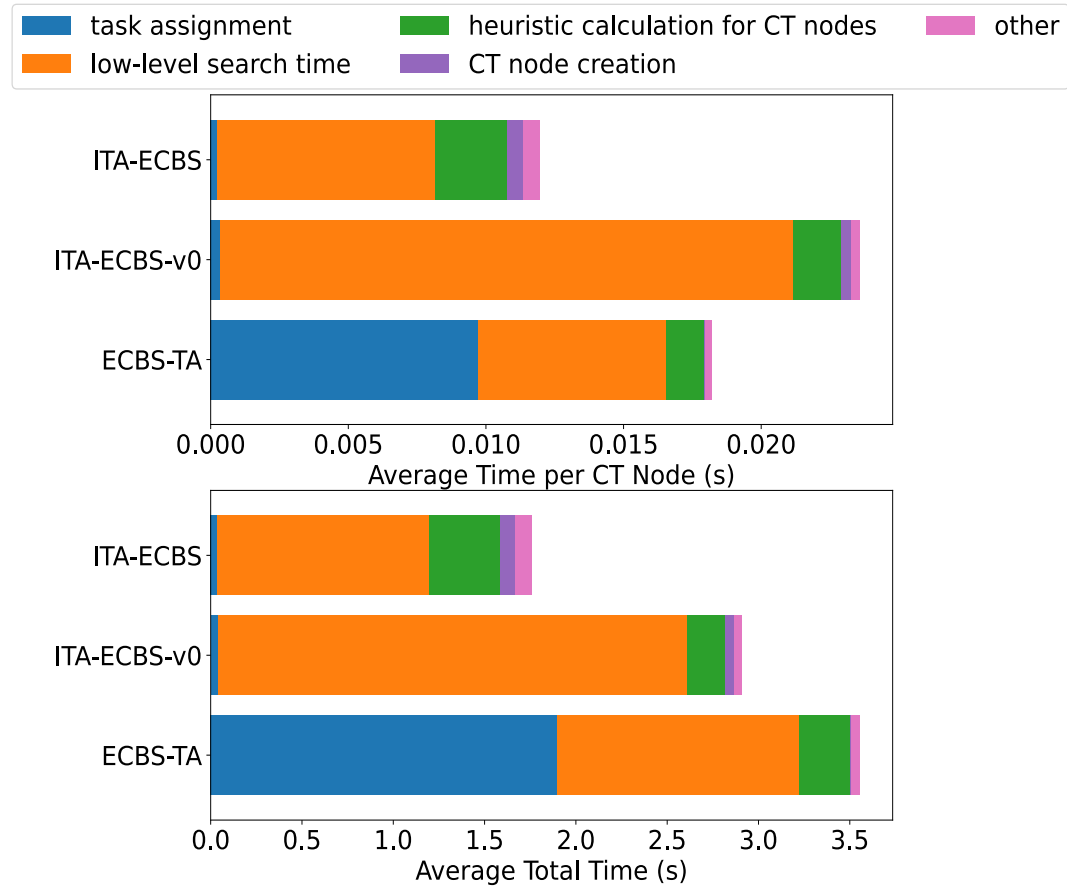
Experiment Results

- ITA-ECBS: two searches (shortest path search + focal search)
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Experiment Results

- two searches (shortest path search + focal search) **faster** than one focal search





Summary

- ITA-ECBS:
 - Bounded-suboptimal method for TAPF



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- We should first get TA solution from LB matrix M_L rather than cost matrix M_C



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- Get LB value first and then search actual path
 - This could be faster than original focal search



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- ITA-ECBS:
 - Bounded-suboptimal method for TAPF
- We should first get TA solution from LB matrix M_L rather than cost matrix M_C
- Get LB value first and then search actual path
 - This could be faster than original focal search
- ITA-ECBS is faster than baseline method in 87% testcases